# A 0D pulsatile model of aortic-arch banding

## Reproducing carotid pulsatility redistribution (Eberth et al., 2009)

svZeroDSolver

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# The problem

**Eberth et al. (2009):** a stiff band on the mouse aortic arch, placed *between* the two carotid takeoffs, creates paired carotids with similar mean pressure & flow but very different pulsatility.

- **RCCA-B** (right) — *upstream* of the band: sees the full pulse + reflections ⇒ pulsatility index rises (PI $\approx 3.11$).
- **LCCA-B** (left) — *downstream*: the resistive–inertial band damps the pulse (PI $\approx 1.65$); baseline CCA $\approx 1.16$.

**Goal:** a reduced-order (0D) lumped model that reproduces this redistribution with the band as the only structural change — and that maps to the wall growth & remodeling Eberth measured.

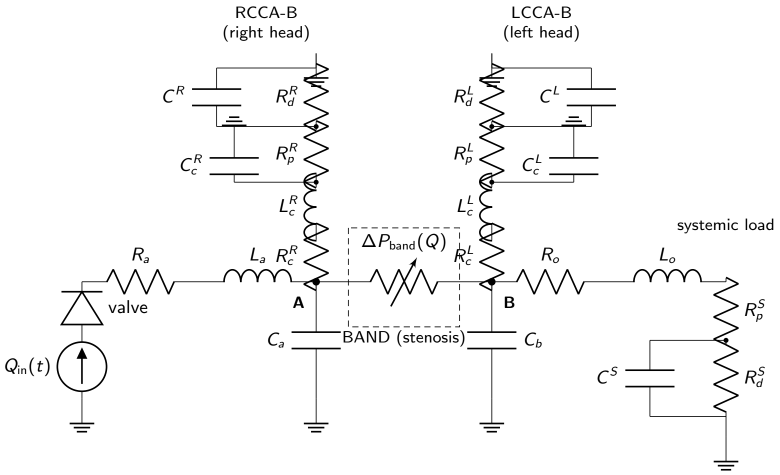
# The 0D model

Heart $\rightarrow$ aortic valve $\rightarrow$ ascending aorta $(R_a, L_a, C_a) \rightarrow$ **node A** $\rightarrow$ band $\Delta P_{\text{band}}(Q) \rightarrow$ **node B** $\rightarrow$ descending aorta $(R_o, L_o, C_b)$; each node feeds a carotid vessel $(R_c, L_c, C_c)$ into a terminal RCR bed $(R_p, C, R_d)$.

- Vessels: RCL `BloodVessel` blocks
- Band: nonlinear *Young–Tsai* stenosis $\Delta P_{\text{band}} = (R_{\text{band}} + S|Q|)Q + L_{\text{band}}\,\dot{Q}$
- Terminal beds: 3-element RCR windkessels
- Valve: `ValveTanh` diode
- Parameters from mouse literature (geometry, blood, wall stiffness $E$)
- Every parameter tagged FIXED / DERIVED / CALIBRATE
- Only the band `stenosis_coefficient` is calibrated (to the Table-1 PI ratio)

Three states differ *only* in carotid geometry: pre-banding (baseline), acute (baseline walls), chronic (Table-1 remodeled walls).

# 0D circuit (electric analogy)

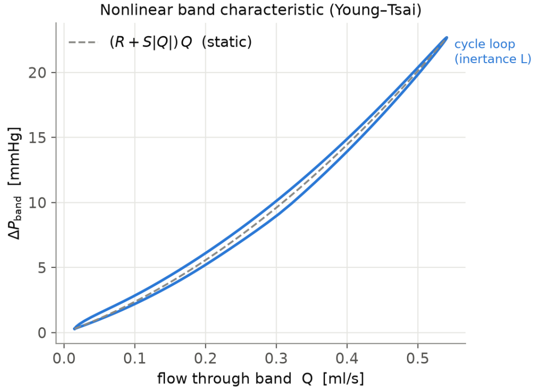



| symbol | component (this repo) |
|---|---|
| $Q_{\text{in}}(t)$ | INFLOW |
| valve | aortic_valve |
| $R_a$, $L_a$, $C_a$ | ascending_aorta |
| $\Delta P_{\text{band}}(Q)$ | aortic_band |
| $R_o$, $L_o$, $C_b$ | descending_aorta |
| node A / B | node_A / node_B |
| $R_c^R$, $L_c^R$, $C_c^R$ | rcca_b |
| $R_p^R$, $C^R$, $R_d^R$ | RCR_RIGHT |
| $R_c^L$, $L_c^L$, $C_c^L$ | lcca_b |
| $R_p^L$, $C^L$, $R_d^L$ | RCR_LEFT |
| $R_p^S$, $C^S$, $R_d^S$ | RCR_SYS |

Not drawn (negligible): aortic_root ($R$, $L$ before the valve). $R_c$, $L_c$, $C_c$ come from each carotid's own Table-1 geometry & $E_{\text{carotid}}$ via Eq. (1)–(3), same as every other vessel. $C_c$ ($\sim$1e-6 ml/mmHg) is $\sim$30–40$\times$ smaller than the bed $C$ it feeds — barely affects PI, so wall remodeling alone doesn't move it (next slide).

Pressure $\leftrightarrow$ voltage, flow $\leftrightarrow$ current. Node A (RCCA-B) is upstream of the band and stays pulsatile; node B (LCCA-B) is downstream and damped. Each terminal bed is an $R_p$–$C$–$R_d$ Windkessel.

# The band is a nonlinear (Young–Tsai) element




$\Delta P$–$Q$ across the band: the quadratic Bernoulli/turbulent term $(S|Q|Q)$ plus an inertance loop $(L\,\dot{Q})$. This nonlinearity + inertia is what damps the downstream pulse while leaving the upstream node pulsatile.

# Parameter roles: fixed, assumed, and the knob we tune

### Fixed (measured / literature)

- blood $\rho = 1.06$, $\mu = 3.5$ cP
- wall $E$: aorta 1.0, carotid 1.5 MPa
- HR 6.09 Hz, CO 0.20 ml/s, MAP 90
- carotid ID/wall (Table 1), aorta dia.
- measured central aortic compliance

$\Rightarrow$ all vessel $R$, $L$, $C$ are *derived* from these via Eq. (1)–(3).

### Assumed (plausible, not fitted)

- segment lengths (carotid, band) — estimates
- ejection waveform shape (systolic fraction)
- RCR split $R_p$:$R_d$ = 10:90; time const. $\tau$
- distal pressure $P_d = 0$
- valve $R_{max}/R_{min}$/steepness

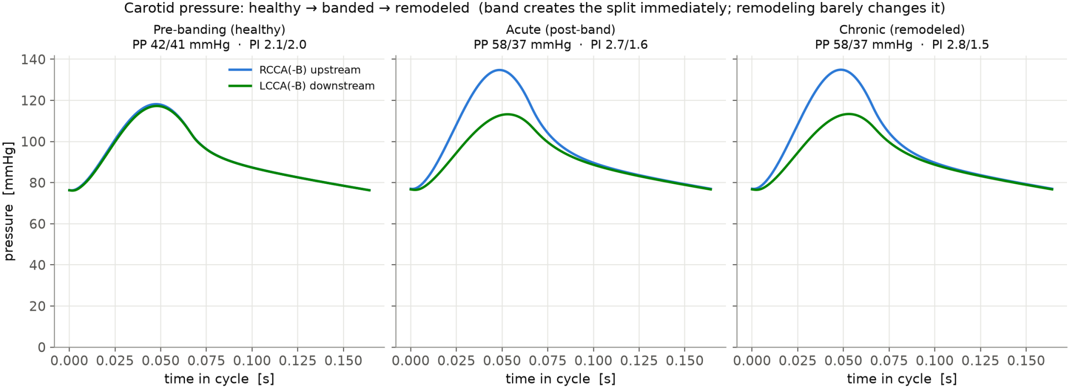
chosen by convention/estimate — could be refined, but not tuned to the data.

### Knob (calibrated to data)

- band `stenosis_coefficient` $S$
- the **one** parameter we turn
- geometry gives a prior; tuned to the Table-1 PI ratio (3.11/1.65)

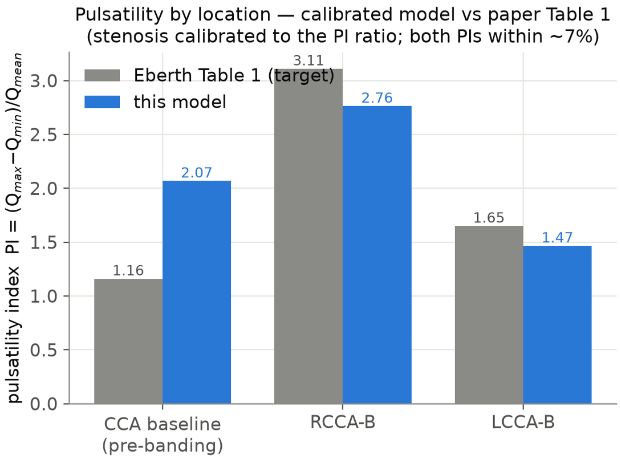
Everything else is fixed or derived — this is the only free knob fitted to the data.

# Result: banding creates the split immediately




Carotid pressure is **symmetric before banding**, splits **immediately** on banding (upstream ↑, downstream damped), and **barely changes** with wall remodeling ⇒ the split is set by band *position*; remodeling is the response, not the cause.

# Result: pulsatility vs. the paper




| | Paper | Model |
|---|---|---|
| RCCA-B PI | 3.11 | $\approx 2.8$ |
| LCCA-B PI | 1.65 | $\approx 1.5$ |
| PI ratio | 1.89 | 1.89 |
| PP RCCA-B | 56 | 58 |
| MAP (R/L) | similar | 98/92 |

The **banded** carotids, the PI ratio, MAP similarity, and pulse pressure are reproduced; mean carotid flows land in the measured range.

# Limitation: baseline vs. banded cannot both be matched in 0D

- The model **over-predicts the baseline (pre-banding) PI** ($\approx 2.1$ vs. the paper's 1.16), though baseline *pulse pressure* matches ($\approx 41$ vs. 42).
- A parameter sweep shows every knob that lowers the baseline PI to 1.16 also collapses the banded PI — the passive bed is a linear pulsatility filter.
- The paper's baseline$\rightarrow$banded amplification is $\sim$2.7$\times$ (1.16$\rightarrow$3.11); a lumped 0D band delivers only $\sim$1.5$\times$.
- The missing $\sim$2$\times$ is **wave reflection** off the band — a spatial/timing effect a 0D element cannot represent.

$\Rightarrow$ Reproducing the *full* amplitude needs a **1D** (wave-propagation) model — exactly the 0D$\rightarrow$1D boundary.

# Targets: pinned, fixed, or an actual free knob?

Control group (CCA) — **1 free knob**

| target | how it's matched | status |
|---|---|---|
| $\bar{Q}$=0.016 | bed $R_p+R_d$ | pinned |
| MAP=92 | sys $R_p^S+R_d^S$ | pinned |
| PP=42 | $C_a$, $C_b$ (aortic) | emergent |
| PI=1.16 | bed $C^{R,L}$ | **KNOB** |

Banded group (RCCA-B / LCCA-B) — **3 free knobs**

| target | how it's matched | status |
|---|---|---|
| $\bar{Q}$=0.022/0.012 | bed $R_p^R+R_d^R$ / $R_p^L+R_d^L$ | pinned |
| MAP=86/77 | sys $R$ (pinned) + band $S$ | **KNOB** |
| PP=56/27 | $C_a$, $C_b$ (aortic) | emergent |
| PI=3.11/1.65 | bed $C^R$ / $C^L$ | **KNOB** |

**Only 4 parameters are actually free/calibrated** (control bed $C^{R,L}$; banded bed $C^R$, $C^L$, band $S$). Bed resistances ($R_p+R_d$, all beds) are *pinned* — solved directly from ($\bar{Q}$, MAP) via Ohm's law, not searched. PP is matched by the aortic $C_a$, $C_b$, which is *fixed* literature identical in both groups — an *emergent* match, not a tuned one. Also fixed: HR (7.17/6.09), CO, carotid geometry, blood $\mu, \rho$, wall moduli $E$, ejection shape.

# How each parameter is actually derived

**1) Bed resistance $R_p + R_d$ — closed form, not searched.** Time-average the RCR equations over one period ($\overline{dP_c/dt} = 0$ in periodic steady state):

$$P_{in} - P_c - R_p Q_{in} = 0, \quad R_d Q_{in} - P_c + P_{ref} - R_d C \dot{P}_c = 0 \quad \xrightarrow{\text{avg}} \quad R_p + R_d = \frac{\text{MAP} - P_{ref}}{\bar{Q}} = \frac{\text{MAP}}{\bar{Q}} \quad (P_{ref} = 0)$$

using each vessel's own measured $\bar{Q}$ and target local MAP; split $R_p : R_d = 10{:}90$ (assumed convention). Same formula for RCR_RIGHT/LEFT/SYS in both groups.

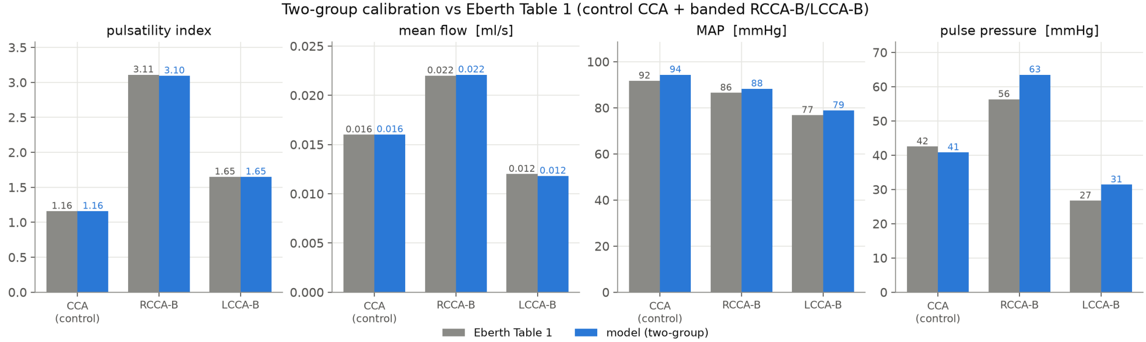
**2) Bed compliance $C$ — bisected against PI (monotone in $C$).** Low $C$: capacitor inert, PI $\to$ PP/MAP (resistive limit); high $C$: capacitor adds flow swing, PI grows. 28-step bisection on $C \in [10^{-7}, 2 \times 10^{-4}]$ until simulated PI $=$ target:

| | target PI | solved $C$ (ml/mmHg) | script |
|---|---|---|---|
| control $C^{R,L}$ | 1.16 | $6.44 \times 10^{-6}$ | `make_control_group.py` |
| banded $C^{R}$ (RCCA-B) | 3.11 | $2.56 \times 10^{-5}$ | `make_banded_group.py` |
| banded $C^{L}$ (LCCA-B) | 1.65 | $2.33 \times 10^{-5}$ | `make_banded_group.py` |

**3) Band $S$ — bisected against the measured A→B pressure drop.** Raising $S$ increases the drop monotonically; bisect $S \in [20, 600]$ until $\text{MAP}_A - \text{MAP}_B = 9.7$ (Table 1) $\Rightarrow S = 95.1$.

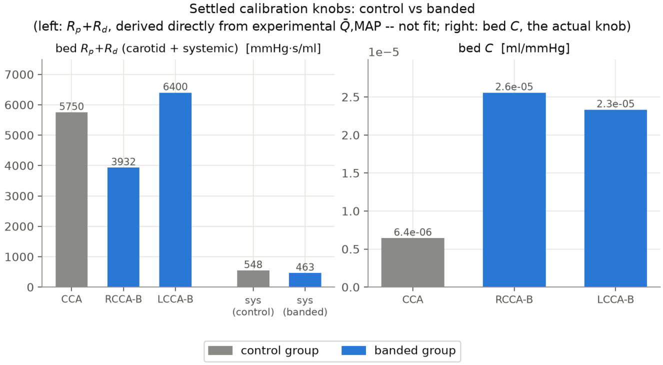
# Revised: two-group calibration (control vs. banded)

Control (CCA) and banded (RCCA-B/LCCA-B) are **two separate animal groups** — they need not share $R/C/L$. Calibrating each on its own terms — HR and measured flows/pressures pin the beds, the band anchored to the measured A→B pressure drop, per-carotid bed compliance sets the PI — matches *all three* carotid states with a stock RCR:



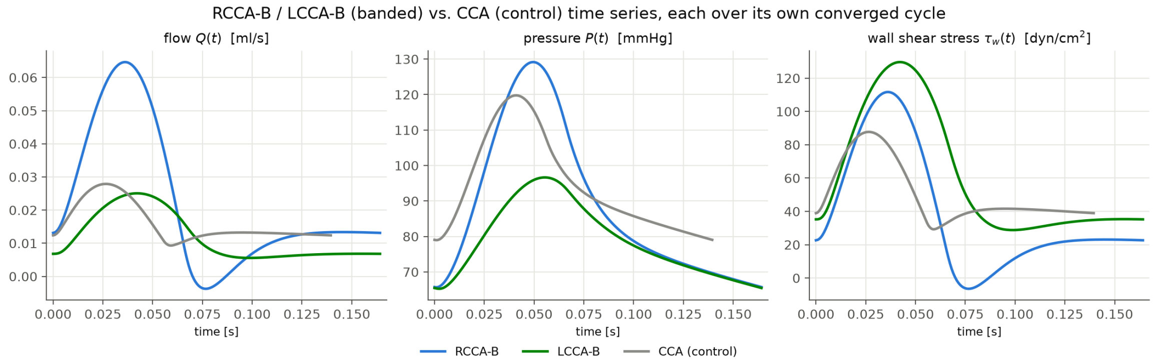

Only **4 parameters are free** (bed $C^{R,L}$, $C^R$, $C^L$, band $S$ — previous slide); bed $R$ pinned, aortic $C$ fixed. Cost: bed compliance differs between groups (fitted) — descriptive, not predictive.

# Settled calibration knobs: control vs. banded




$R_p$+$R_d$ (left, both carotid and systemic beds) is **derived directly from experimental data** ($\bar{Q}$, MAP via Ohm's law) — not fit. Bed $C$ (right) is the one actual calibrated knob.

# Looking ahead: inputs for growth & remodeling (G&R)




Flow $Q(t)$ and pressure $P(t)$ come directly from the solver; wall shear stress $\tau_w(t)$ is **not** a solver output — computed via Poiseuille $\tau_w = 4\mu Q(t)/(\pi r^3)$ using each vessel's fixed lumen radius. These per-cycle time series (not just the mean/PI summaries used for calibration) are the raw inputs a future G&R model would consume.

# Summary

- A literature-parameterized 0D model reproduces the Eberth **pulsatility redistribution**: upstream carotid pulsatile, downstream damped, at matched mean pressure/flow.
- Banding creates the split **immediately**; wall remodeling barely changes it — consistent with pulsatility being the *stimulus* for remodeling.
- With a *single shared* parameter set, 0D matches the redistribution and pressures but not both the baseline and banded PIs (a bounded amplification limit; a 1D wave-reflection effect is small for the mouse).
- Treating control and banded as *separate groups* — their own beds, band anchored to the measured pressure drop — matches **all three carotid states**, at the cost of group-specific (fitted) bed compliance.